

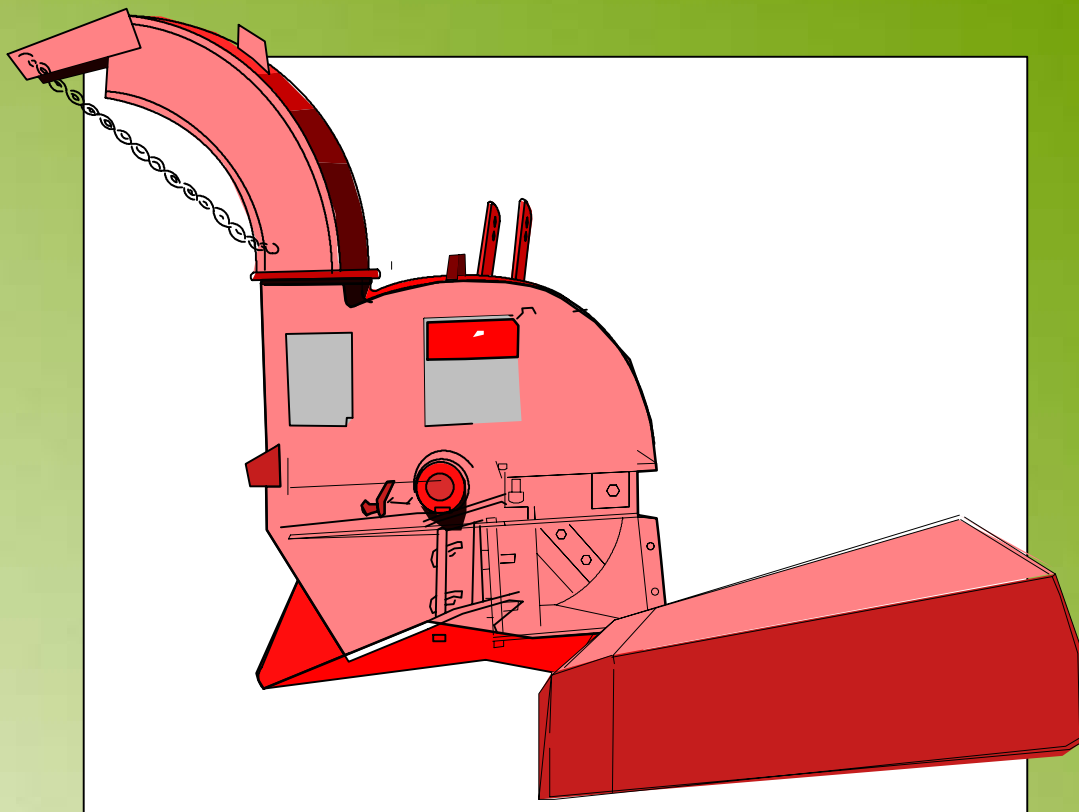


CH140 L

OPERATION, MAINTENANCE AND SPARE PARTS

VALBY

CHIPPER CH140 L



From machine: 334 5011

It is very important to read this instruction handbook thoroughly before using the machine!

Farmi Forest Corporation

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CH140 L

1. TO THE OWNER – OPERATOR

Thank you for purchasing a Farmi Forest Corporation product.

Before we begin this manual, the people at Normet Corporation want you, the owner-operator, to know this piece of equipment can be very dangerous if safety procedures and warnings are ignored. Read this manual carefully page by page until you understand it completely. Failure to do so will result in personal injury and / or equipment damage.

All personnel including operators, maintenance crew, etc. should read this manual before start up.

This manual should be considered a permanent part of your machine and should remain with the machine when you sell it.

IMPORTER:

**NORTHEAST IMPLEMENT Co.
P.O. Box 402
Spencer, N.Y. 14883
U.S.A**

Tel.	+1 – 607 – 589 6160
Fax	+1 – 607 – 589 4026
E – mail	valby@baka.com

2.TERMS AND SYMBOLS USED IN INSTRUCTIONS

**THINK
SAFETY**

Safety is a combination of operator common sense and alertness at all times when the machine is being operated or serviced.



This message is used for general reminders of good safety practices or to direct attention to unsafe practices. The message will appear in your operator's manual and/or the sign will appear on the machine with the color combination of yellow and black.



This message denotes a specific potential hazard. The sign will be displayed on the machine in areas of potential hazard. The sign will have the color combination of yellow and black.



This message denotes the most serious specific potential hazard. The sign will be displayed on the machine in areas of potential hazard.



Messages labeled "Important" will appear in your operator's manual to provide specific instructions for performing adjustments, services, etc. If these instructions are not followed it could result in damage to the machine.

CH140 L

3.SAFETY

3.1.GENERAL SAFETY



Do not operate this machine until you have read and understood the manual page by page.

The owner of this machine is responsible for instructing all operators and support personnel in the operation and safety precautions of this chipper. Proper training prior to operating the chipper is obligatory.

- Inform everyone who works with the chipper about the risks and how they can avoid accidents.
- Before the chipper is running, ensure that the working place is clear of any bystander within a minimum risk zone of 100 ft. (30 meter).
- Do not let children or untrained persons operate the chipper.
- Always have the chipper mounted on the 3–point hitch of the tractor. Otherwise the chipper will tip over.
- There is a serious crushing hazard between tractor and chipper. Do not go between the tractor and the chipper in any case, for example when you are mounting the chipper to the tractor.
- Check that the covers of the universal shaft are in their place and the transmission chains are fastened.
- Safety goggles, earmuffs and a hard hat are obligatory when chipping. Use respirator mask, if necessary.
- Do not wear loose clothing, loose sleeves, scarves or long uncovered hair around moving parts of machinery. Also avoid gloves of poor condition or with loose fitting, because they might get caught in the branches.
- Stay alert! Do not operate the chipper when fatigued, or under the influence of alcohol or drugs.
- **IMPORTANT!** Never leave the chipper running and unattended.
- **IMPORTANT!** Failure to obey the warnings on the chipper or in the operator's manual will result in personal injury and / or death and / or equipment damage.
- Check the material before you feed it. The material must be free from nails, stones and all other materials that are not wood, paper or plastic. **IMPORTANT!** Feeding wire and barbed wire into the chipper is very dangerous as it might pull in the operator.
- Make sure that no one is in the path of flying chips. Point the discharge pipe away from windows, doorways and other areas where people or animals may be.
- Keep the work area clean so there is nothing to trip over.
- Disengage the PTO, cut the engine and remove the keys before any maintenance.
- Use only original parts replacements. **IMPORTANT!** Do not make any modifications.
- Check all moving parts. They should be fastened in place, in good working condition and all shields and guards must be in place.
- Point the discharge downwind to prevent dust or chips from blowing onto the operator.
- If a wood jams inside the chipper, cut the engine and wait for all movements to stop before reaching inside the feed chute.
- Do not use the chipper indoors when you use it with a the tractor or with the separate hydraulic motor. It's possible to use the chipper indoors with the electric motor use.
- You must be very careful when working indoors. Protect yourself against possible hazards like exhaust gas by sufficient ventilation, for dust that can catch fire by cleaning your environment regularly.
- Always use the lock bolt before and during maintenance on the machine.
- Place the PTO shaft in the holding device when disconnected from tractor.



Risk of cutting injury or hits by flying debris.

Do not reach into the feed chute for any reason.

Failure to follow these instructions could result in serious injury or death.

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3.2.SAFETY INSTRUCTIONS FOR THE HYDRAULIC FEEDER AND FEED CHUTE

- IMPORTANT! Always also read the general safety instructions.
- IMPORTANT! Do not ever put any part of your body inside the feed chute!
- Prior to operating get familiar with the function and controls of the machine. All functions must be in blameless condition.
- Test the stop function.
- Before starting up, check that there is no debris or foreign objects in the feed chute.
- Do not work in front of the feed chute opening! Stand on the left side of the chute when feeding, as the feed roller might jerk the wood up or to the right.
- When checking malfunction, all controls must be in "off" or in "stop" position.
- Let go of the wood when the feed roller or the knives get hold of it. To feed short pieces, push them in with longer ones.

3.3.SAFETY INSTRUCTIONS FOR THE HYDRAULIC UNIT

- Before performing lubrication work on any equipment, consult the manufacturer's instruction manual and follow the recommended procedures.
- Before lubricating the machine, shut down the engine and stop the machine.
- Pressurized oil can be hazardous. Pressurized oil jet can penetrate skin and cause serious injuries or death.
- Occasional skin exposure with oil is not dangerous. If though you can not prevent long-time exposure use protective gloves and other protective clothing.
- Keep flammable material away from heat, sparks and open fire.
- Hot oil and oil spray can be hazardous. Small amounts swallowed or inhaled do not usually cause toxication. Old oil is more dangerous than new.
- If oil has got into eyes flush them with cool water after which go to doctor. Use safety goggles.
- Avoid mixing lubricants of different grade and quality. Use only recommended oils or oils with corresponding characters.

3.4.SAFETY INSTRUCTIONS FOR THE SEPARATE HYDRAULIC MOTOR USE

- Check the moving parts. They should be fastened in place, in good working condition and all shields and guards must be in place.
- Inform everyone who works with the chipper about the risks and how they can avoid accidents.
- Do not let children or incapable persons operate the chipper. Do not wear loose clothing, loose sleeves, neckties or long uncovered hair around moving parts of machinery.
- Disengage the PTO and turn the tractor off before you service the chipper.
- Use only original parts replacements. Do not make any modifications.
- Failure to obey the warnings printed on the chipper or in the operator's manual might result in a serious injury or death.
- When checking the malfunction, all devices must be in "off" position.
- Place the PTO shaft in the holding device when disconnected from tractor.

3.5.PARKING THE CHIPPER

- Ensure that the chipper is placed on a hard level and horizontal surface.
- Ensure that there is no material in the chipper or in the feed chute except for the remainder branch.
- Clean the working area.

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WHEN ORDERING SPARE PARTS, PLEASE INDICATE MACHINES TYPE FROM THE MACHINE PLATE, SPARE – PARTS ORDER NUMBER, DESCRIPTION AND QUANTITY REQUIRED.

EXAMPLE: CH140 L, 43510220, Knife, 2 pcs.

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4.CHIPPER PRESENTATION

4.1.FIELD OF APPLICATION

CH140 is a multi purpose chipper for processing small-sized wood up to Ø 6,3" (160 mm). It produces uniform chips to be used as fuel. CH140 is a mobile, light-weight machine for handling all kinds of wood waste from parks, roadsides and other environmental cleaning. The chipper can be fitted with a separate hydraulic power feed.

4.2.CHIPPER ACCESSORIES

- Single or dual motor hydraulic feed, manual or gravity hopper
- Independent hydraulic unit HD11
- Separate hydraulic motor drive HD 100
- Skid Steer adaptation
- Transmission T160

4.3.MAIN COMPONENTS

1. *UPPER CHAMBER*
2. *LOWER CHAMBER*
3. *DISC*
4. *KNIVES*
5. *VERTICAL ANVIL*
6. *HORIZONTAL ANVIL*
7. *TWIG BLADE*
8. *DISCHARGE CHUTE*
9. *LID*
10. *HOLDING DEVICE FOR THE PTO SHAFT*

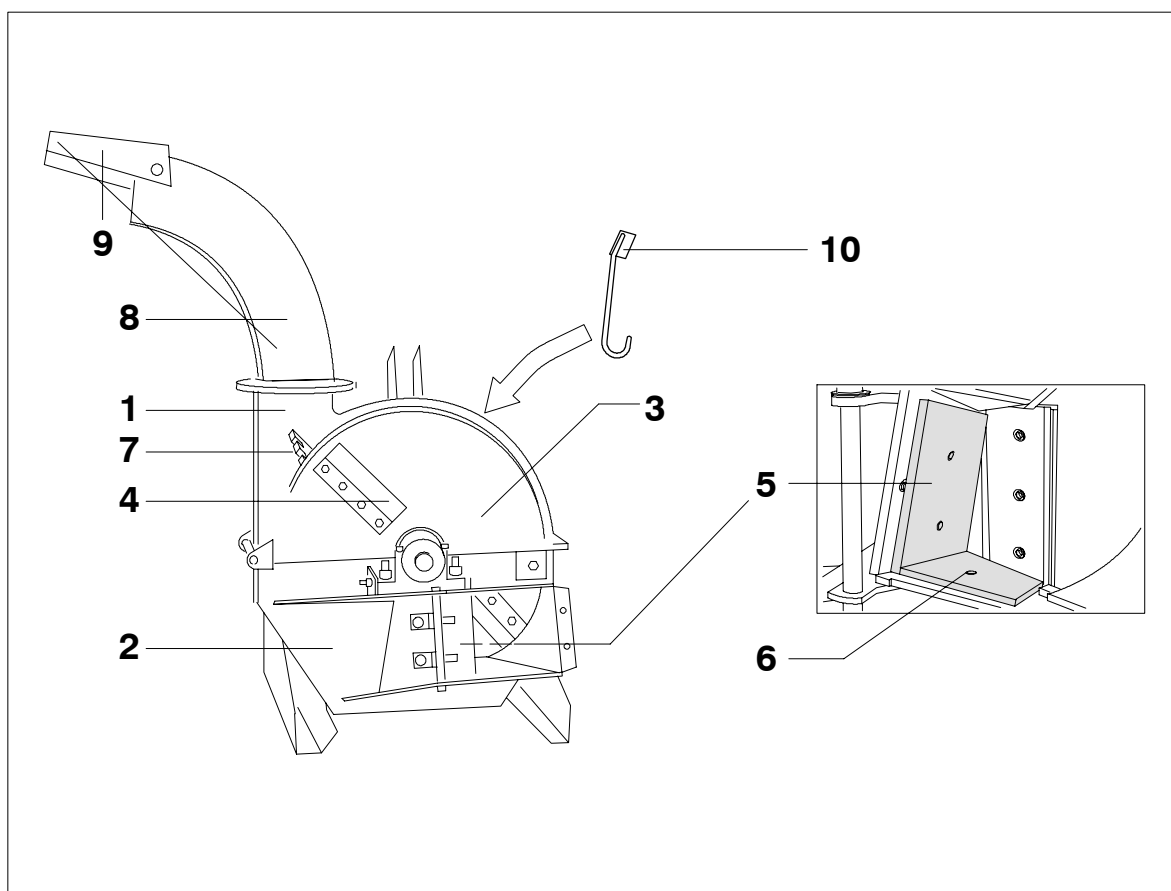


Fig.1.Main parts.

4.4.DIRECTIVES AND STANDARDS APPLIED

4.4.1.UNIFIED STANDARDS

- EN292-2
- EN294
- EN60204-1

4.4.2.NATIONAL STANDARDS

- PrEN13252
- prEN1553-1

CH140 L

5. TECHNICAL DATA

Output	5–20 m ³ /h (177–706 ft ³ /h)
Mean chip length	13 mm (1/2")
Max wood diameter	160 mm (4 1/2")
Power demand	10–40 kW (15–40 H.P.)
Tractor PTO speed	540–1000 RPM
Number of knives	2 pcs.
Power source	Tractor or hydraulic motor
Mounting	3–point hitch CAT I
Rotor diameter	Ø 670 mm (26 1/3")
Rotor weight	100 kg (220 lbs)
Rotor's rotation speed	900–1700 RPM
Bearings	Spherical roller bearings
Discharge chute	360° adjustable
Discharge chute height	86" (1,2 m)
Opening of upper chamber	Single hinge
Chipper weight (no feeder)	625 lbs (285 kg)
Oil demand, P.T.O. speed – chip 13 mm (1/2"), 540 rpm	3–10 gal / min
Type of feed chute	Manual chute or hydraulic feed
Sound pressure level	102 dB (A)
Sound power level	120 dB (A)
CEN / TC144 WG8N16	

5.1. DIMENSIONS

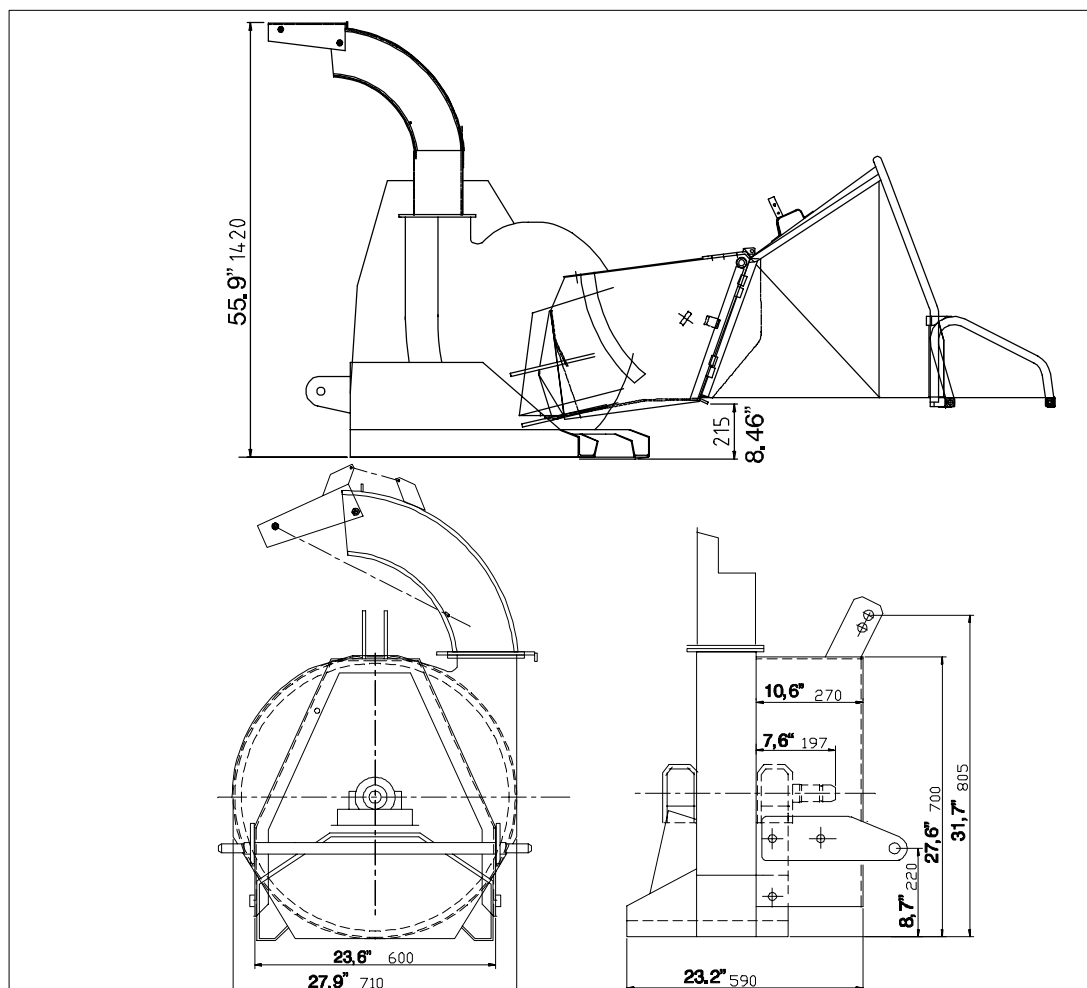


Fig.2. Dimensions of CH140.

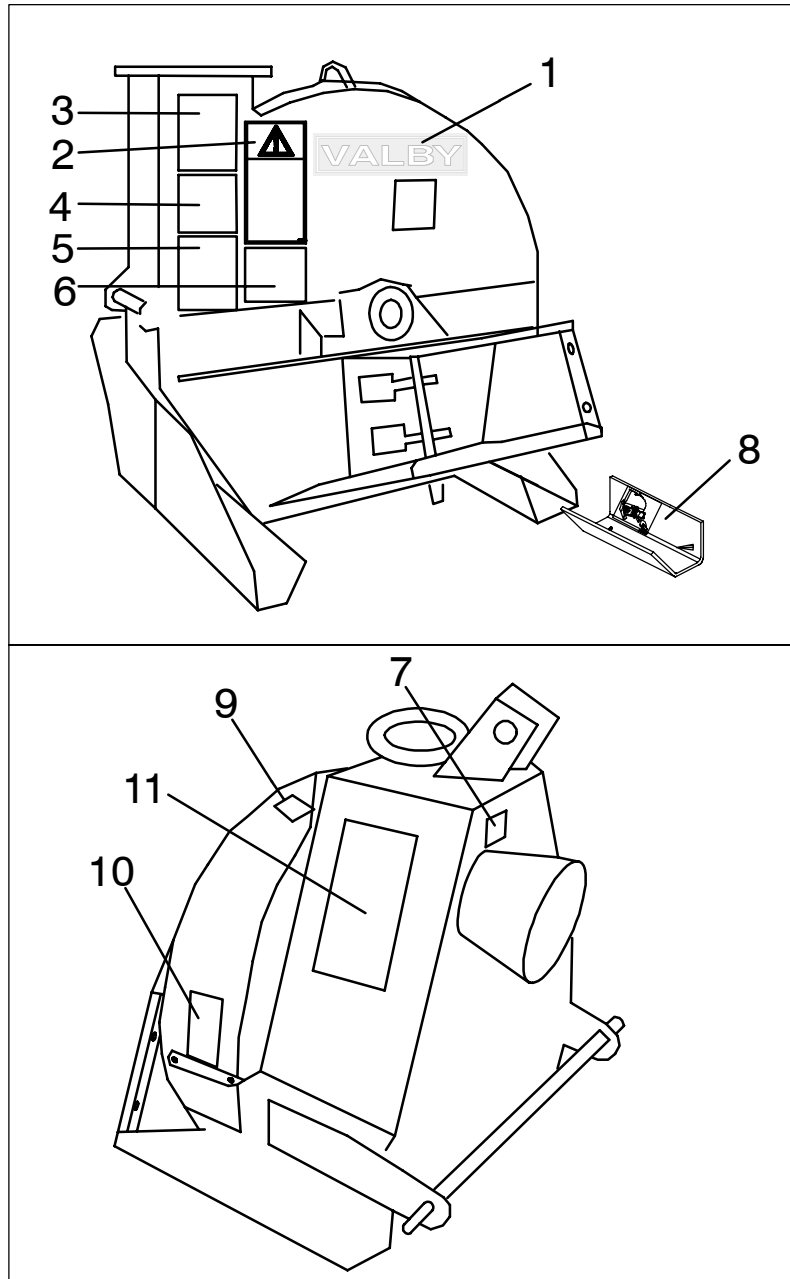
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5.2.PLATES AND STICKERS FOR THE CHIPPER

IMPORTANT!



Do not remove any plate or sticker placed by the manufacturer on the chipper or the accessories. Replace missing ones immediately. You can order them as a spare parts.



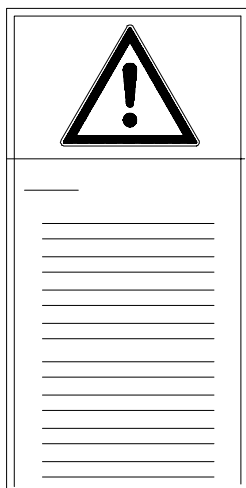
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1.



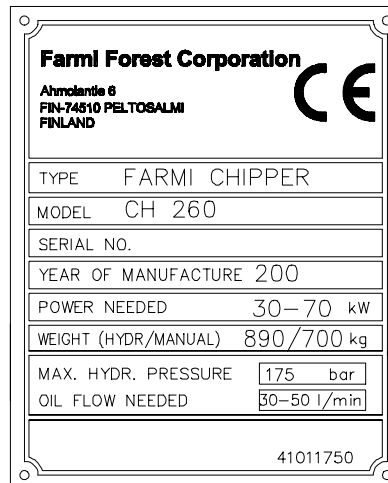
1 pc 30140602

2.



1 pc 41014310

3.



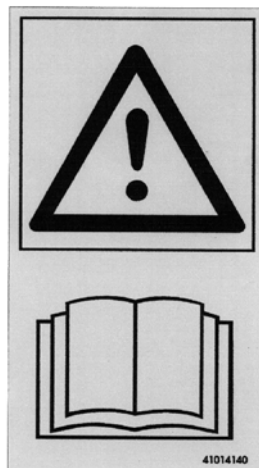
1 pc 41011750

4.



3 pc 40142080

5.



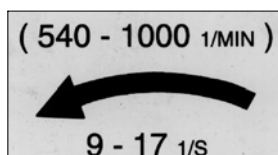
1 pc 41014140

6.



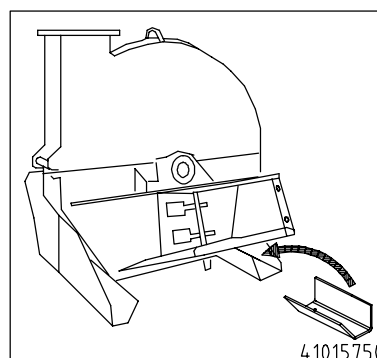
1 pc 41014170

7.



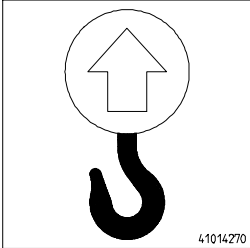
1 pc 40141160

8.



1 pc 41015750

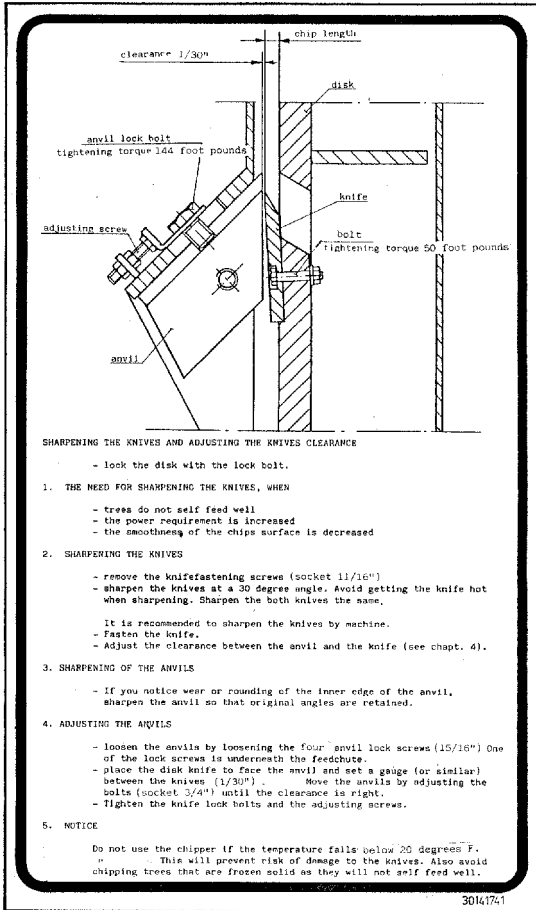
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9.  41014270

5 pc 41014270

10.  41014110

2 pc 41014110

11.  30141147

1 pc 30141147

43513680 Plates and stickers for chipper CH 140

Part	Order no.	Description	Qty.
1. ...	30140602	VALBY-sticker	1
2. ...	41014310	Safety precautions	1
3. ...	41011750	Machine plate	1
4. ...	40142080	Safety goggles, earmuff and a hard hat are obligatory when chipping	3
5. ...	41014140	Do not operate this machine until the owner's manual has been completely read and understood	1
6. ...	41014120	Safety precautions before maintaining the chipper	1
7. ...	40141160	Rotation speed and rotation direction of the PTO shaft	1
8. ...	41015750	Mounting the extension plate	1
9. ...	41014270	Lifting point	1
10. ...	41014110	Beware of cutting tools	2
11. ...	30141147	Sharpening the knives	1

6. TRAFFIC ON PUBLIC ROADS

IMPORTANT!



Prior to every procedure always recall the general safety instructions given in chapter 3.

Follow these special instructions to avoid risks of accidents. Otherwise the manufacturer or retailer will not assume any liability for damage.

You shall have full control of the unit in every situation when driving.

Bumpy driving might damage the chipper / the feeding device.

Always follow the regulations established by the law.

6.1. TRANSPORT POSITION

1. Open the pawl lever of the feed chute and turn the chute aside. Disconnect one end of the control lever from the valve and lock the chute into transport position.
2. Turn the discharge chute into transport position.
3. Before driving off, ensure the machine unit's appropriate stability and check that the chipper is firmly connected to the tractor and all fasteners are in place.

IMPORTANT!



Make sure that the rear lights of the tractor are visible. Use a separate light panel, if needed.

7. PREPARATIONS FOR START UP



WARNING!

Do not attempt to operate this chipper without proper training. Read and understand the owner's manual before operating the machine.

Failure to heed the warnings printed on the chipper or in the operators manual might result in serious injury or death!

7.1. LIFTING

IMPORTANT!



Prior to every procedure always recall the general safety instructions given in chapter 3.



DANGER!

Make sure that nobody will enter the lifting zone.



Always check before lifting that the lifting equipment is in order and use the marked lifting points, whenever possible. Always use proper lifting equipment and check for its sufficient lifting capacity.

Know the weight of the load and check that the performance of the lifting equipment will not be exceeded.

Check the hoisting cables and chains regularly. Mark any discarded cable and remove it immediately.

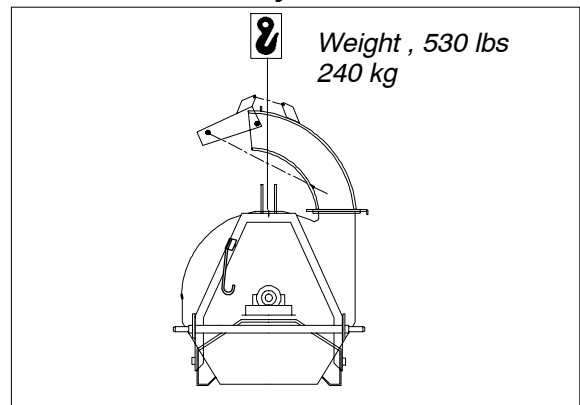


Fig.3. CH170 lifting point.

7.2. MOUNTING THE CHIPPER UNIT

1. Mount the 3-point hitch lift bar to the chipper frame.
2. Tighten the nuts with a 1 3/16" or 30 mm wrench to 190 foot-pounds (260 Nm).
3. Mount the feeding device to the chipper with a hinge pin and, from the other side, with two M12x40 bolts and M12 nuts. See fig. 4. (If you use a hydraulic feeder, refer to separate manual for instructions on mounting and connecting the hydraulics.)
4. Connect the discharge chute to the chipper with three M10 bolts and nuts. Adjust the lid control with the rod.
5. A Power Take-Off shaft (PTO shaft) transfers the power to the chipper. The chipper has a 1 3/8" diameter 6 spline PTO hookup.
6. Check for the correct length of the PTO shaft. (Refer to 7.4.)

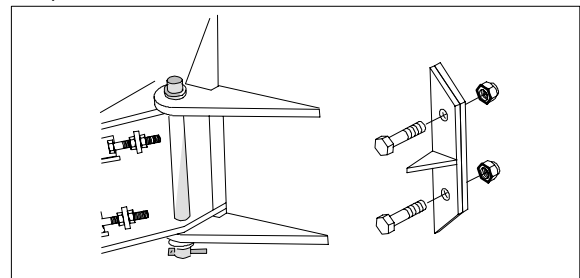


Fig.4.

- Turn off the engine.
- Lower the chipper on its legs on a hard, level surface.
- Turn the chipper's axle by hand to ensure that the disk can turn freely and that there are no foreign objects in the chipper.

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- Look inside feedchute to ensure that it is empty.
- Make sure that all shields are in place and intact. Removal of the shields is prohibited!
- Direct the discharge chute in desired position.

7.3.MOUNTING THE EXTENSION PLATE

- If better stability is needed, use the extension plate. Look also spare parts.

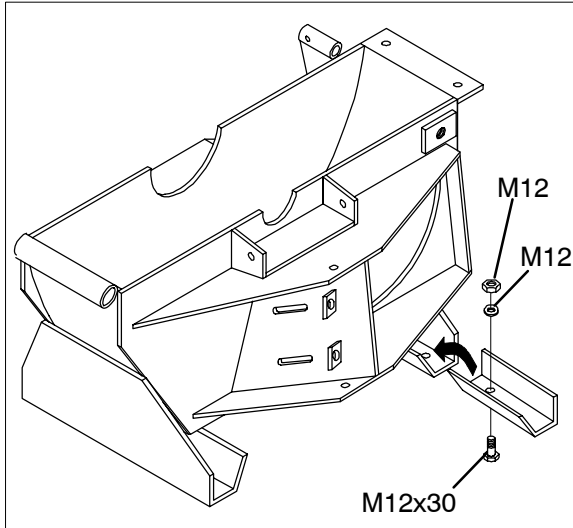


Fig.5.Mounting the extension plate.

7.4.PTO–SHAFT



If the tractor has a PTO–brake (for example Ford), you have to use an overrunning clutch. Shaft recommendations: M4SLC

7.4.1.CUTTING THE PTO SHAFT TO THE CORRECT LENGHT



If the PTO shaft is too long it might bind when the three point hitch is lifted up. Damage might be caused to the bearings of the chipper or to the PTO of the tractor. If the PTO shaft is too long it must be cut. Both PTO halves must be shortened by equal amounts.

1. Mount the chipper on the three–point hitch of the tractor.
2. Cut the engine and remove the keys. Lower the chipper on a firm and level surface.
3. Check that the splined shafts of the tractor and chipper are on the same level.
4. Connect one shaft half to the chipper PTO and the other half to the tractor PTO. Put the PTO halves side by side so that one shaft end is 1" from the PTO–end of the chipper. Mark the length required. The tubes must be so short that they cannot reach the bottom in any chipper position. Cut the plastic tube. Refer to fig. 6.–7.

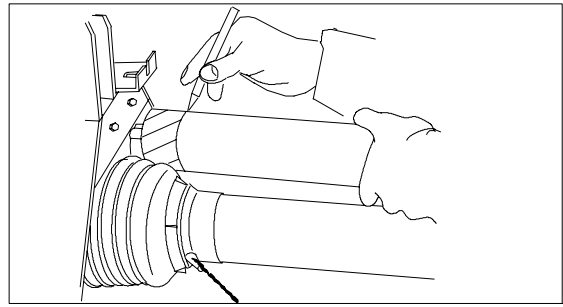


Fig.6.

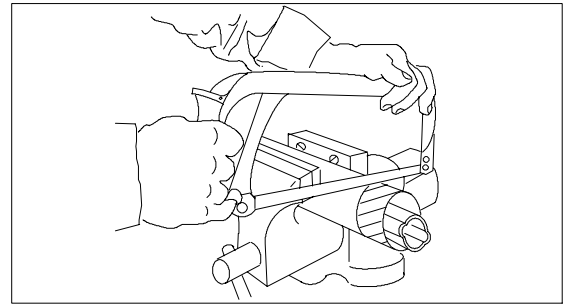


Fig.7.

5. Cut a profile tube piece of similar length as the plastic tube piece. Shorten the other PTO shaft similarly. Finish by filing the sharp edges. See fig. 8.– 9.

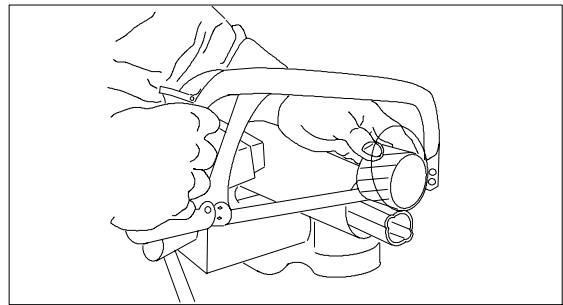


Fig.8.

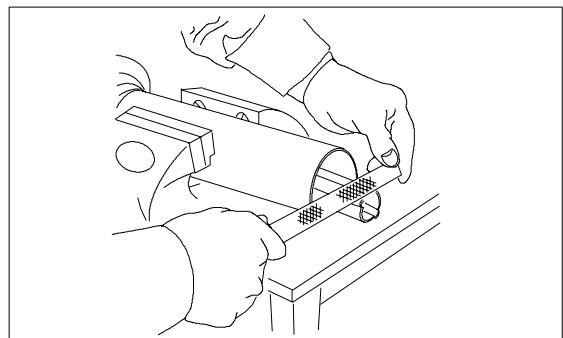


Fig.9.

8. OPERATING THE CHIPPER

8.1. SAFETY WHEN OPERATING



IMPORTANT!

Prior to any procedure also recall the general safety instructions in chapter 3.



CAUTION!

Inform everyone who works with the chipper about the risks and how they can avoid accidents.



DANGER!

Never leave the chipper running and unattended.

Do not ever stand between the tractor and the chipper.

Before the chipper is running, ensure that the working place is clear of any bystander within a minimum risk zone of 50 ft. (15 meter).

Do not let children or untrained persons operate the chipper.

Do not use the chipper disconnected from the power source.

When using a tractor, always have the chipper mounted on the 3-point hitch of the tractor. Otherwise the chipper will tip over.

Check that the covers of the PTO shaft are in their place and the transmission chains are fastened.

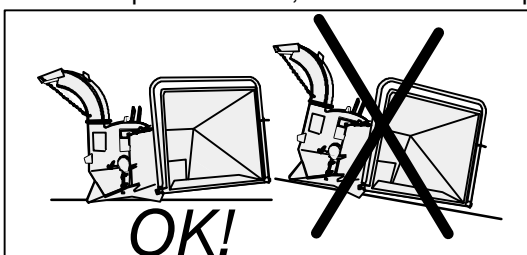
Safety goggles, earmuffs and a hard hat are obligatory when chipping. Use respirator mask, if necessary.

Do not wear loose clothing, loose sleeves, scarves or long uncovered hair around moving parts of machinery. Also avoid gloves of poor condition or with loose fitting, because they might get caught in the branches.

Stay alert! Do not operate the chipper when fatigued or under the influence of alcohol / drugs.

8.2. PRIOR TO OPERATING

1. Disengage the PTO, cut the engine and remove the keys.
2. Lower the chipper on its legs on a hard, even ground as shown in picture below, so that it will not tip over.



3. Check for the free rotation of the disk by turning the chipper axle by hand. Also check for any foreign object in the chipper and the feed chute.
4. Make sure that all covers are in place and intact. **Do not ever remove any cover!**
5. Direct the discharge tube into desired position.



IMPORTANT!

The PTO shaft of a new chipper needs lubricating prior to use!



IMPORTANT!

When using a new chipper, always check the tightness of bolts and fasteners after one operating hour.

8.3. STARTING THE CHIPPER

- **IMPORTANT!** Whenever engaging the PTO clutch on your tractor, you are turning a flywheel with a load of 300-600 lbs. Engage the PTO slowly allowing the flywheel to start to rotate. Once you have the chipper running, you may completely engage the PTO and then bring up the rpm's to operating speed (540 - 1000 rpm).

8.4. CHIPPING



IMPORTANT!

Prior to any procedure also recall the general safety instructions in chapter 3.



DANGER!

Beware of rotating knives. Risk of cutting injury.

Feeding wire and barbed wire into the chipper might drag in the operator.

Do not reach inside the feed chute until all movements have stopped!



CAUTION!

Always keep proper fire fighting equipment on hand when chipping. Keep an eye on the outside temperature of the chipper to prevent overheating of the knives. Stop chipping and let the chipper cool down.

Good maintenance and a tidy working place is of utmost importance for preventing fire. Check for hot bearings.

Pour water down the feed chute, if the chipper starts smoking.

Safety goggles, earmuffs and a hard hat are obligatory when chipping. Use respirator mask, if necessary.

Do not wear loose clothing, loose sleeves, scarves or long uncovered hair around moving parts of machinery. Also avoid gloves of poor condition or with loose fitting, because they might get caught in the branches.

Fig.10. Choose a level ground.

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If a wood jams inside the chipper, cut the engine, remove the keys and wait for all movements to stop before reaching inside the feed chute.

Look out for flying debris!

Before feeding, make sure that the material is free from stones, nails and all other materials that are not wood, paper or plastic. Also avoid chipping wood that are frozen solid, as they will not self-feed well.

Do not stand in front of the feed chute when feeding. The feed roller might jerk the wood up or to the right.

- Feed the wood standing on the left side of the chute.
- When chipping wood, push the wood trunk inside the feed chute until the knives or the feed roller touches the wood. The chipper is self-feeding, so let go of the wood as soon as the knives or the feed roller touches it.
- Point the discharge downwind to prevent dust or chips from blowing onto the operator. Point the discharge pipe away from windows, doorways and other areas where people or animals may be.
- Keep the work area clean so there is nothing to trip over.
- If you are using a hydraulic feeder, refer to separate manual.



Do not use the chipper in temperatures below -5°F (-20°C). Risk of knife damage!

8.5. REDUCING THE POWER REQUIRED.

When using a small tractor to power the CH 140 chipper, select a 540 rpm PTO speed. Run the tractor at full engine speed and the PTO will turn about 600–650 rpm.

Should the tractor stall when chipping, reduce the power demand by making the chipper cut with only one knife: Remove one knife and attach it to the other side of the disk. Now you will be able to chip material twice as thick. **IMPORTANT! Feed the material at half speed.**

1. Disengage the PTO, cut the engine and remove the keys.
2. Open the upper housing bolts (3/4" or 19mm wrench) and turn the upper housing aside.
3. Lock the disk with the lock bolt, fig. 20.
4. Turn the feed chute to the side.
5. Use protective gloves when handling the knives.
6. First remove the lock nut and then the knife using a 17 mm (11/16") socket wrench. The disk bolts are threaded through the disk so the bolts must be turned out.
7. Before fastening the knife on the other side, prepare the bolts as follows: Place a spring washer on the bolt. Screw on the lock nut and tighten it to 45 Nm (38 foot-pounds).
8. Place another spring washer on the bolt and thread the bolt through the knife into the disk as shown in figure 11. Tighten the bolts to 45 Nm (38 foot-pounds). **IMPORTANT! To retain the balance always use all four fastening bolts of the knife.**



Interchange the cutting and counterweight knives at times to ensure that they are equally much worn and sharpened, and the disk stays in balance.

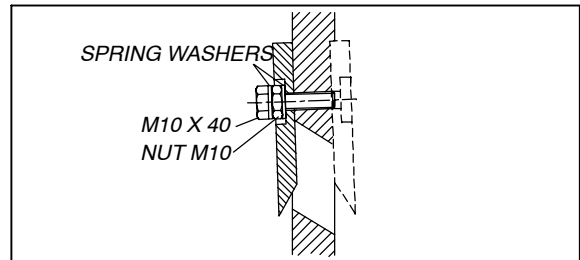


Fig. 11. Remove the knife and attach it to the other side of the disk. The dotted line shows the knife in its normal cutting position.

8.6. PROCEDURES AFTER OPERATING

8.6.1. CLEARING THE CHIPPER

- Prior to shutting down the chipper, always run a large branch of $\varnothing$ 80 to 100 mm (3 to 4") into the knives and let the chipper cut off a few inches. By then reversing the feed direction the knives will be cleared.



To eliminate the possibility of jamming the chipper when starting up again, leave a branch in the chipper as a reminder of having cleared the knives.

8.6.2. TURNING THE CHIPPER OFF IN TRACTOR USE

- Let the tractor rpm idle before turning the PTO to OFF-position. This is especially important on tractors with PTO brakes, such as Ford, where you, after idling the tractor, should slowly move the PTO control lever to the OFF-position.

8.6.3. TURNING THE CHIPPER OFF IN HYDRAULIC MOTOR USE



DANGER!

After turning the chipper off wait for all movements to stop; the disk will continue to rotate by flywheel momentum after the PTO has been disconnected.



Never disconnect the hydraulics from high rpm's. This will cause risk of cavitation and might damage the hydraulic motor. Check that no sounds of cavitation emit from the hydraulic motor.



To prevent cavitation it is very important to slowly reduce the base machine rpm's to idle before disconnecting the hydraulics.

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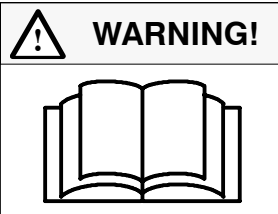
9.SAFETY MAINTENANCE AND SERVICE



Prior to any procedure also recall the general safety instructions in chapter 3.

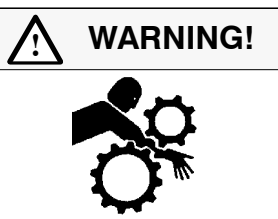
Use only original spare parts.

Observe common safety practice.



Before performing lubrication work on any equipment, consult the manufacturer's instruction manual and follow the recommended procedures.

Ensure that the chipper is on a firm and level surface.



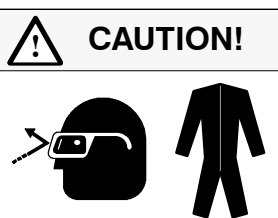
Before lubricating the machine, shut down the engine and stop the machine.

When handling the knives, use protective gloves.

Do not ever stand between the tractor and the chipper. Modifications of chipper are prohibited.



Pressurized oil can be hazardous. Pressurized oil jet can penetrate skin and cause serious injuries or death.



Occasional skin exposure with oil is not dangerous. If though you can not prevent long-time exposure use protective gloves and other protective clothing.



Keep flammable material away from heat, sparks and open fire.

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9.1.BEFORE LUBRICATION AND MAINTENANCE



Always disengage the PTO and turn the tractor OFF before you service or repair the chipper.

Wait for all movement to stop before reaching inside the feed chute.

Remove the keys so that the tractor can not be started up accidentally.

Lock the disk for service or repair

Park the chipper on hard and level ground, so it will not fall down.

9.2.LUBRICATION



Hot oil and oil spray is hazardous.

Avoid skin contact with oil and grease. Always consult a physician in cases of getting oil in the eyes or possible skin changes.

Avoid mixing lubricants of different grade and quality. Use only recommended oils or oils with corresponding characters.

Follow the instructions and regulations of the manufacturers and the authorities.

Always wear proper clothing, skin protecting crèmes or appropriate gloves.

Never use lubricating oils or greases to clean your hand with. Metal particles and additives in the lubricants might damage the skin.

9.2.1.LUBRICATING THE BEARINGS

The bearings are factory –greased with Shell Alvania Grease R 3. Use a similar lubricant, such as Kendall L427. **IMPORTANT! Too much grease will cause heat, which reduces lubrication.**

- Lubricate the bearings every 200 working hours or once a year.

1.Disengage the PTO, cut the engine and remove the keys.

2.Open the upper housing bolt and remove / turn aside the upper housing.

3.Lock the disk with the lock bolt, fig. 20.

4.Remove / turn aside the feed chute.

5.Remove the top of the bearing housing. Remove as much old grease as you can and replace it with new, but do not fill the housing completely. Refer to Replacing the bearing as advised in chapter 9.3.1.

6.Replace the top of the bearing housing. Tighten to 38 foot–pounds (50 Nm).

9.2.2.LUBRICATING THE PTO SHAFT

- Lubricate the PTO shaft prior to operating and with intervals shown in fig. 12.

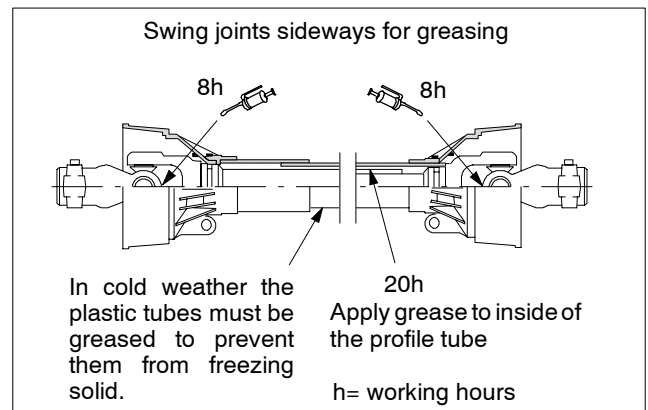


Fig.12.The lubrication spots and intervals of the P.T.O shaft.

9.3. BEARINGS

9.3.1. REPLACING THE BEARINGS

7. Remove the fastening bolts M10 (7) and the upper bearing housings (6).
8. Mark the location of the tightening cone on the shaft.
9. Lift the disk.
10. Bend the claw of the securing ring (3) out from the notch on the axle nut and open the axle nut.
11. Remove the axle nut, securing ring, spacer ring (4), bearing (5), spacer ring (9), and tightening cone (10).
12. Install the tightening cone, spacer ring (9), and bearing on the disk shaft. Note the thickness of the spacer ring (9) – see Fig. 13.
13. Install the securing ring with the claws facing outwards and the inside claw in the groove of the tightening cone, and install the axle nut.
14. The inner ring of the bearing should press tightly against the tightening cone.
15. Tighten the axle nut with a hook spanner until the bearing is tightly on the cone, or to 80 Nm ($58 \frac{15}{16}$ lbf). However, the outer ring of the bearing should turn freely. Note the location of the tightening cone on the shaft.
16. Bend one claw of the securing ring (3) into a notch on the axle nut.
17. Install the other half of the dust cover and end plate on the bearing housing. Install the spacer ring (4) on the shaft.
18. Lower the disk to the bearing housing.
19. Attach the other half of the dust cover to the upper bearing housing; install the upper bearing housing fastening bolts and tighten to 50 Nm ($36 \frac{7}{8}$ lbf).
20. Lubricate the bearing housing. An excessive amount of grease causes overheating and impairs lubrication.

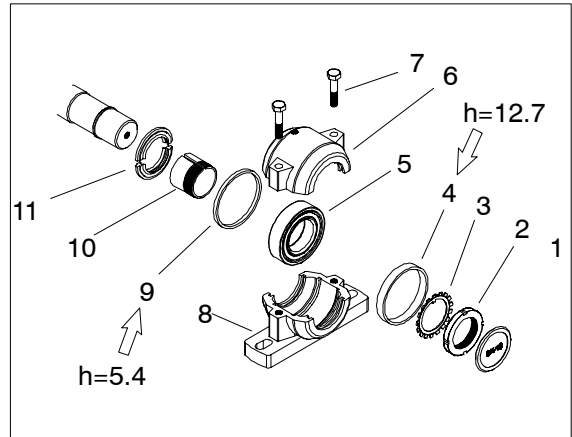


Fig. 13. Bearing, feeder side.

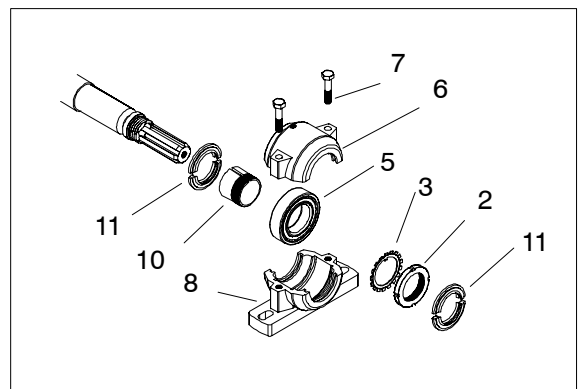


Fig. 14. Bearing, splined shaft side.

- The bearings at the feeder side and splined shaft side are different from each other. The bearing at the splined shaft side, Fig. 14., does not have spacer rings (4 and 9) and an end plate (1). The bearing at the splined shaft side has dust covers on both sides (11).

9.3.2.ADJUSTING THE BEARING CLEARANCE

1. Remove the fastening bolts (M10) (7) and the upper bearing housings (6).
2. Remove the grease from the bearing housing.
3. Lift the disk.
4. Bend the claw of the securing ring (3) out from the notch on the axle nut and open the axle nut.
5. Remove the spacer ring (3) if the bearing is at the feeder side.
6. Measure the radial clearance on top of the bearing between the rollers and outer ring with a feeler gauge. The clearance should be 0.02–0.03 mm (0,0008–0,0012 "). Measure the clearance by pushing the feeler gauge between the rollers (point A, Fig. 16.) through the bearing and then moving the gauge back and forth between the rollers and outer ring – see Fig. 17. Do not force the feeler gauge through the clearance.
7. If the clearance exceeds 0.03 mm (0,0012"), bend the claw of the securing ring (2) out from the notch on the axle nut (1).
8. The bearing is tightened by turning the axle nut clockwise with a 70 mm or 2 3/4" hook spanner until the right clearance is achieved. Do not tighten by hammering the axle nut.
9. Turn the axle nut clockwise until the notch is aligned with the nearest claw of the securing ring. Bend the claw into the notch. Do not bend the claw that was bent earlier.
10. Lower the disk to the bearing housing.
11. Install the upper bearing housing and tighten the bolts (6) to 50 Nm (36 7/8 lbf).
12. Lubricate the bearing housing. An excessive amount of grease causes overheating and impairs lubrication.

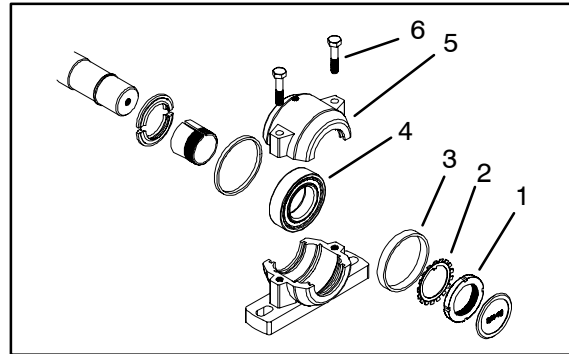


Fig. 15. Bearing, feeder side.

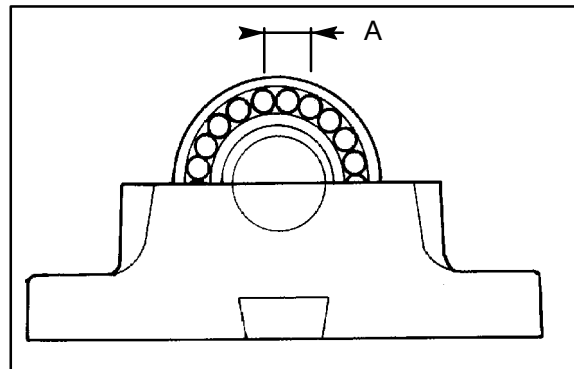


Fig. 16. Push the feeler gauge between the rollers and outer ring.

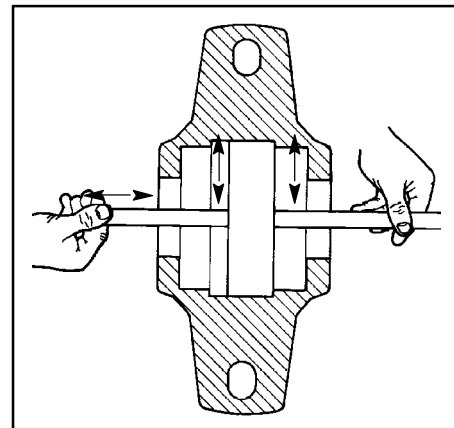


Fig. 17. Measuring the clearance.

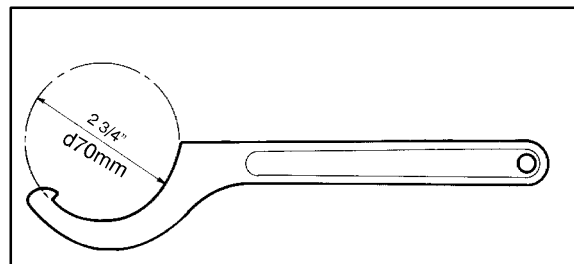


Fig. 18. Tightening the bearing

9.4.CHANGING THE SPLINED AXLE

IMPORTANT! Only a qualified person is allowed to change the splined axle.

- 1.Disengage the PTO, cut the engine and remove the keys.
- 2.Open the upper housing bolts and remove it.
- 3.Remove / turn aside the feed chute.
- 4.Remove the top of the bearing housings.
- 5.Lift away the disk.
- 6.Remove the bearings, fig. 13.

IMPORTANT! Welding damages the bearings.

7.Grind a 7/16" (11 mm) deep groove about 11/16" (17 mm) from the edge of the axle, fig.19.

8.Turn the axle so that it comes loose and can be removed.

9.Clean the hole before mounting the new axle. Tap the new axle slightly in place.

10.Protect the joint faces from splashes before welding.

11.Weld a fillet weld, for example ESAB 68.81, OK 48, OK Femax 38.65 with 3 runs. Finish the welding.

12.Reassemble the chipper.

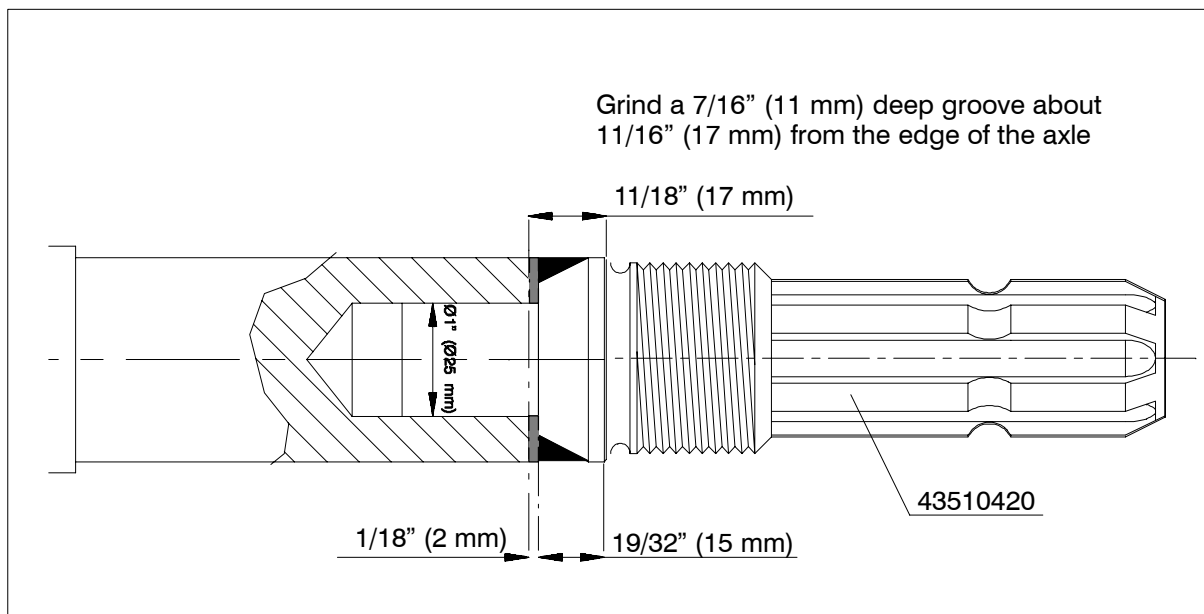


Fig.19.Changing the splined axle.

9.5.KNIVES AND ANVILS



ROTATING KNIVES!
Wait for all movement to stop before reaching inside the feed chute.

The disk continues rotating by flywheel momentum after the PTO has been disengaged.

Use protective gloves when handling the knives.

Failure to follow safety precautions could result in serious personal injury or death.

9.5.1.SHARPENING THE KNIVES

The knives need sharpening when:

- The wood does not self-feed well.
- The power demand has increased.
- The chip surface is rough.



By keeping the knives as original pairs they will wear equally and the disk will stay in balance.

Avoid getting the knife hot when sharpening. Sharpen both knives the same amount.

- 1.Disengage the PTO, cut the engine and remove the keys.
- 2.Open the upper housing bolts (3/4" or 19mm wrench). Turn the upper housing to the side.
- 3.Lock the disk with the lock bolt, fig. 20.
- 4.Remove or turn the feed chute to the side.
- 5.Remove the four locknuts behind the knife frame (socket size 11/16" or 17mm). See fig 20.
- 6.Remove the fastening bolts of the knife, which are threaded through the disk (socket size 17mm or 11/16"). Be aware of knuckle injury, if the wrench should slip when turning.

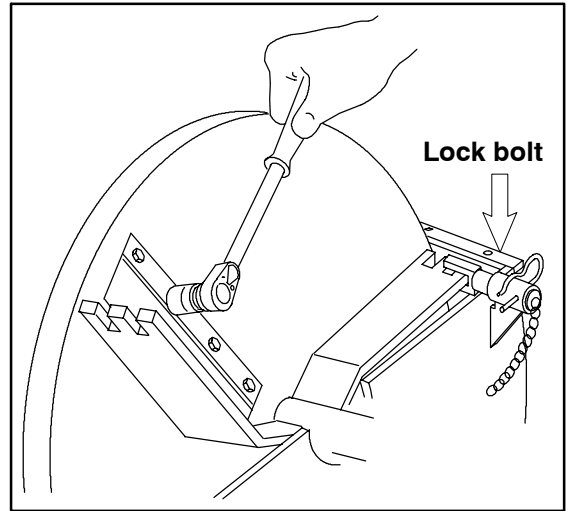


Fig.20.

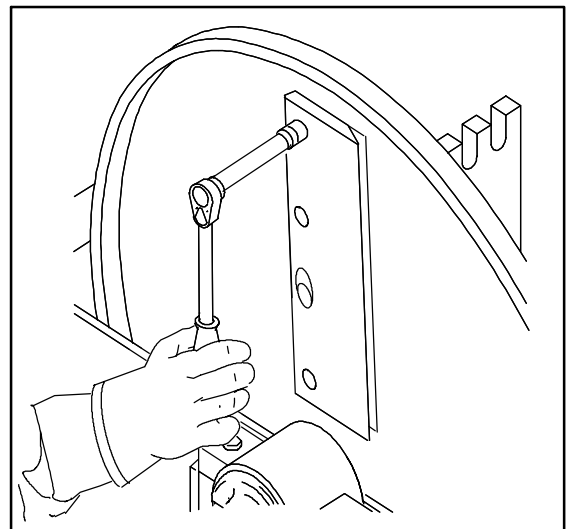


Fig.21.

- 7.Sharpen the knives at an angle of 30 degrees. Refer to fig. 22.

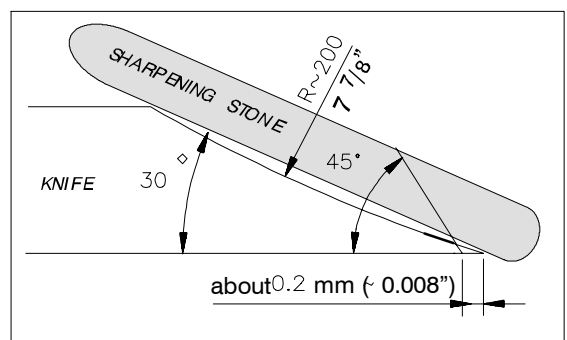


Fig.22.

- 8.Check the condition of fastening bolts and nuts. Clean the knife pockets before reinstalling the knives.
- 9.Fasten the knife and tighten the bolts to 38 foot-pounds or 45 Nm.
- 10.If needed, adjust the anvil clearance as advised in the next chapter. The anvil clearance has to be adjusted

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after every 5 to 10 sharpening, when the clearance between the knife and anvil exceeds 1,5 mm or 1/17".

11.Reassemble

9.5.2.ADJUSTING THE CLEARANCE BETWEEN THE KNIVES AND THE HORIZONTAL ANVIL

Normally you can sharpen the knives 10 times before the anvil needs to be adjusted. Adjust the anvil if the clearance to the edge of the knives exceeds 1,5 mm or 1/16 of an inch. Refer to fig. 24.

- 1.Disengage the PTO, cut the engine and remove the keys.
- 2.Open the upper housing bolts (3/4" or 19mm wrench). Turn the upper housing to the side.
- 3.Lock the disk with the lock bolt, fig. 20.
- 4.Remove or turn the feed chute to the side.
- 5.Loosen the three fastening bolts (A, B) of the anvil. One of them is located under the feed chute. The wrench size is 15/16" or 24 mm.
- 6.Turn the disk so that a knife and the anvil are opposite each other. Adjust the anvil with the nuts (C.) fig. 23. wrench size 3/4" or 19 mm) until a clearance of 1/20...1/30 of an inch. If you do not have a gauge, use the tickness of a matchbox cover as a guide.
- 7.Tighten the adjusting nuts (C). Tighten the anvil fastening bolts (A, B) to 200 Nm or 155 foot-pounds.

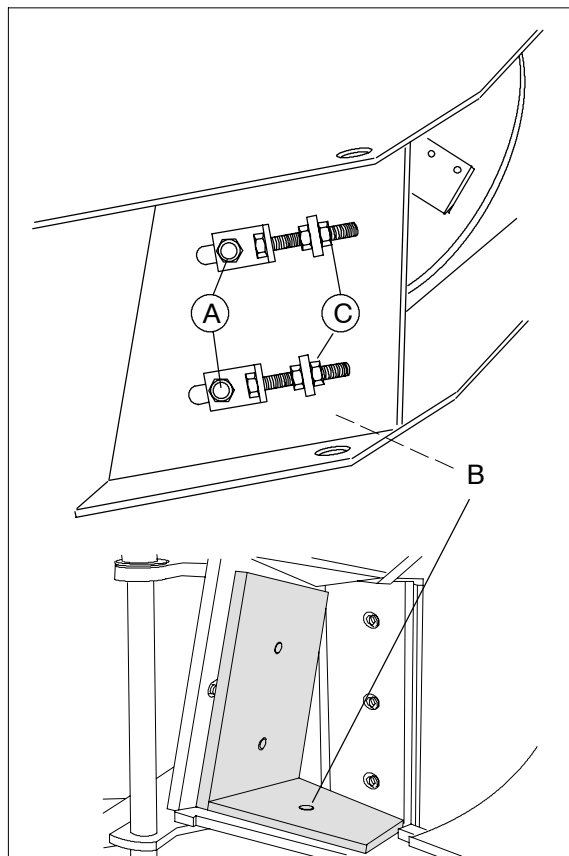


Fig.23.

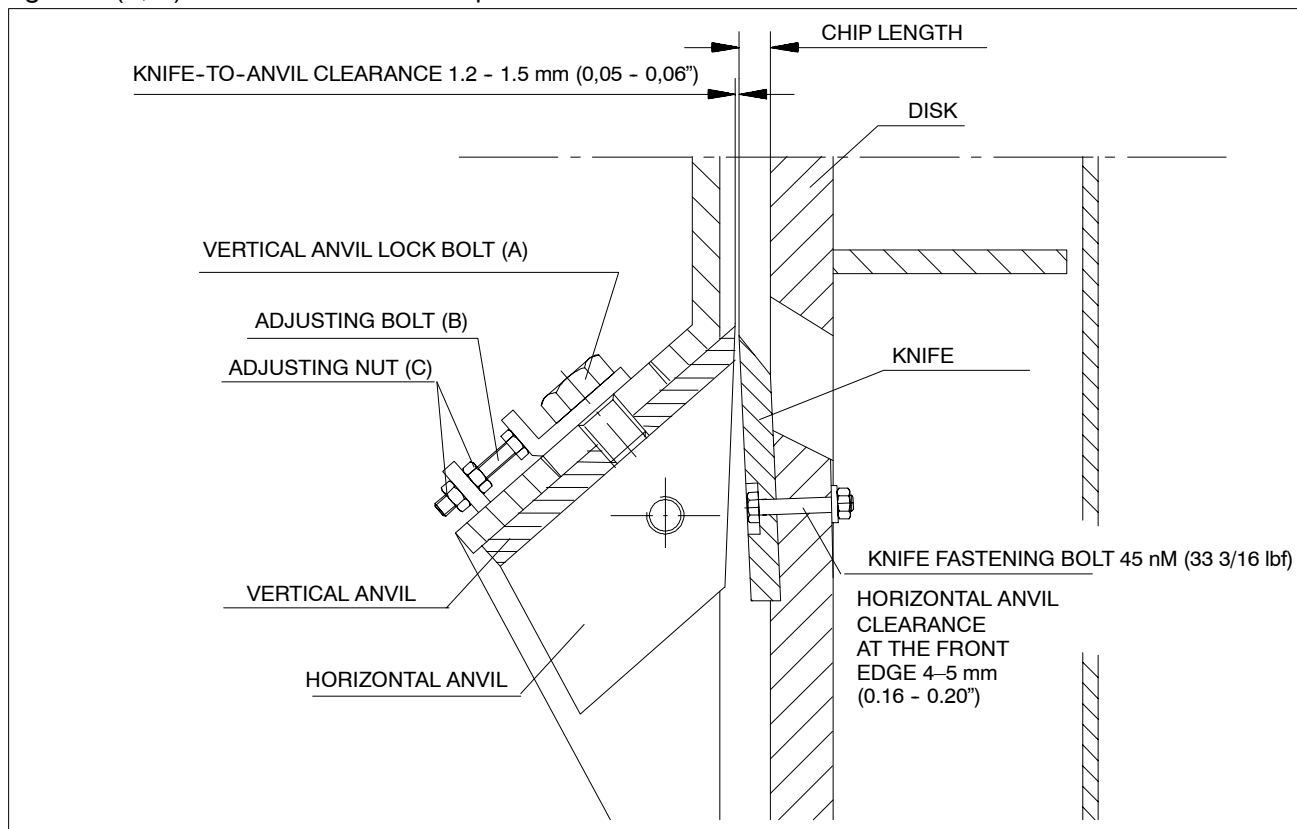


Fig.24.

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9.5.3.SHARPENING THE ANVIL

If you notice wear or rounding of the inner edge of the anvil, sharpen the anvil so that original angles are retained.

9.6.TIGHTENING THE BOLTS

- The tightness of fasteners and the bolts in the chipper must be checked once a week. **IMPORTANT! On a new machine check the tightening after one hour of use.**
- Refer to Maintenance and lubrication schedule, chapter NO TAG Tighten as indicated in the specifications below.

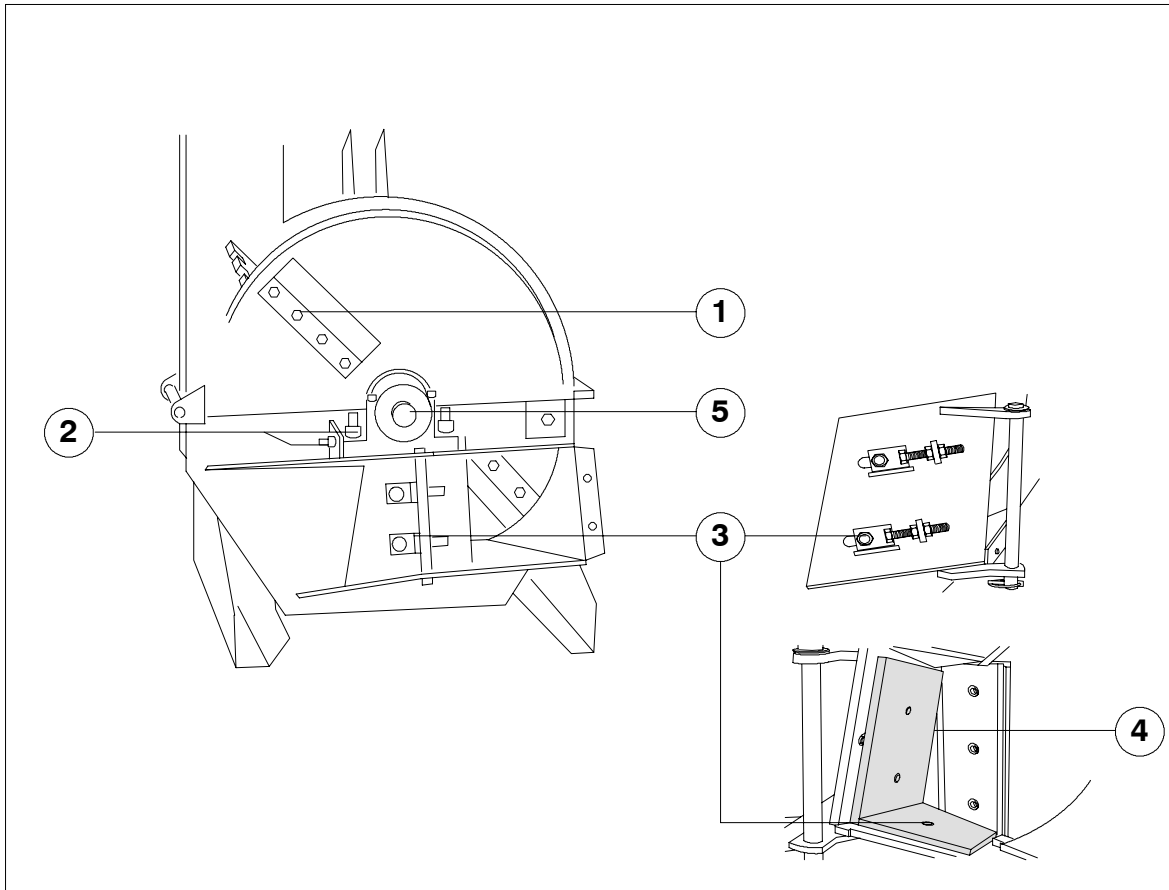


Fig.25..

9.6.1.TORQUES AND CLEARANCES

ITEM	Socket size	Torque, Nm (lbf)
1, Check knife bolts tightness	17mm (11/16")	60 (44 ³ / ₁₆)
2, Check bearing housing bolts tightness on both sides	19mm (3/4")	80 (58 ¹⁵ / ₁₆)
3, Check anvil bolts tightness	24mm (15/16")	200 (147 ³ / ₈)

ITEM	Socket size	Measurement
4, Check clearance between knives and anvil	19mm (3/4")	1,2-1,5 mm (0.05 - 0.06")
5. Check bearing's radial clearance		0.02-0.03 mm (0.0008" to 0.0012")

10.STORING AND DISCARDING

10.1. STORING THE CHIPPER

- If the chipper is to be stored for a longer period, cover the knives with grease and make sure that the water drains of the lower housing are open.

10.2. DISCARDING THE CHIPPER

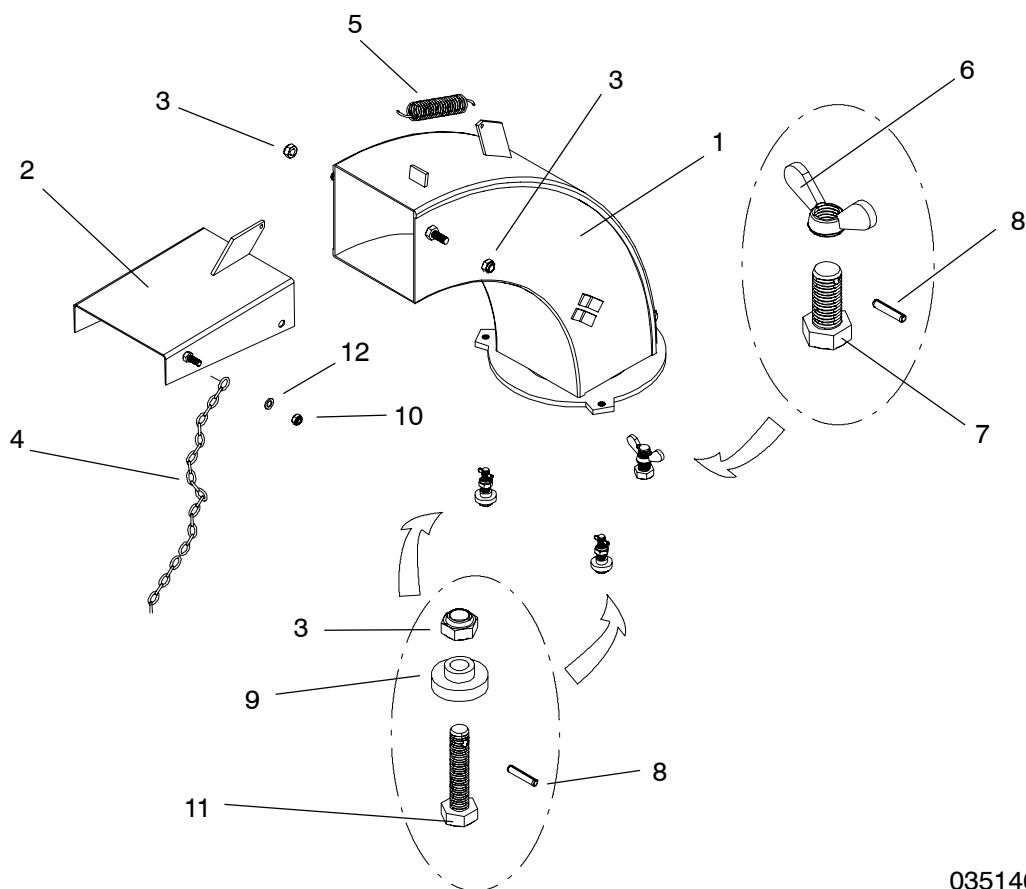
- When the machine comes to the end of its working life, it has to be duly discarded. Contact local authorities for more information.

11.CRIME PREVENTION

- Contact local authorities and / or your retailer.
- If a theft occurs, notify the law enforcement agency having jurisdiction and the insurance carrier immediately.
- Give a full description and a complete set of serial and identification numbers to the investigating officer and insurance carrier.
- If available, provide the investigating officer with photographs of the actual machine, manu–facturers literature and knowledge of any iden–tifiable marks that would assist in identifying the machine.
- Regularly check the identification plates on all machines and report any missing or destroyed plate.

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Discharge pipe

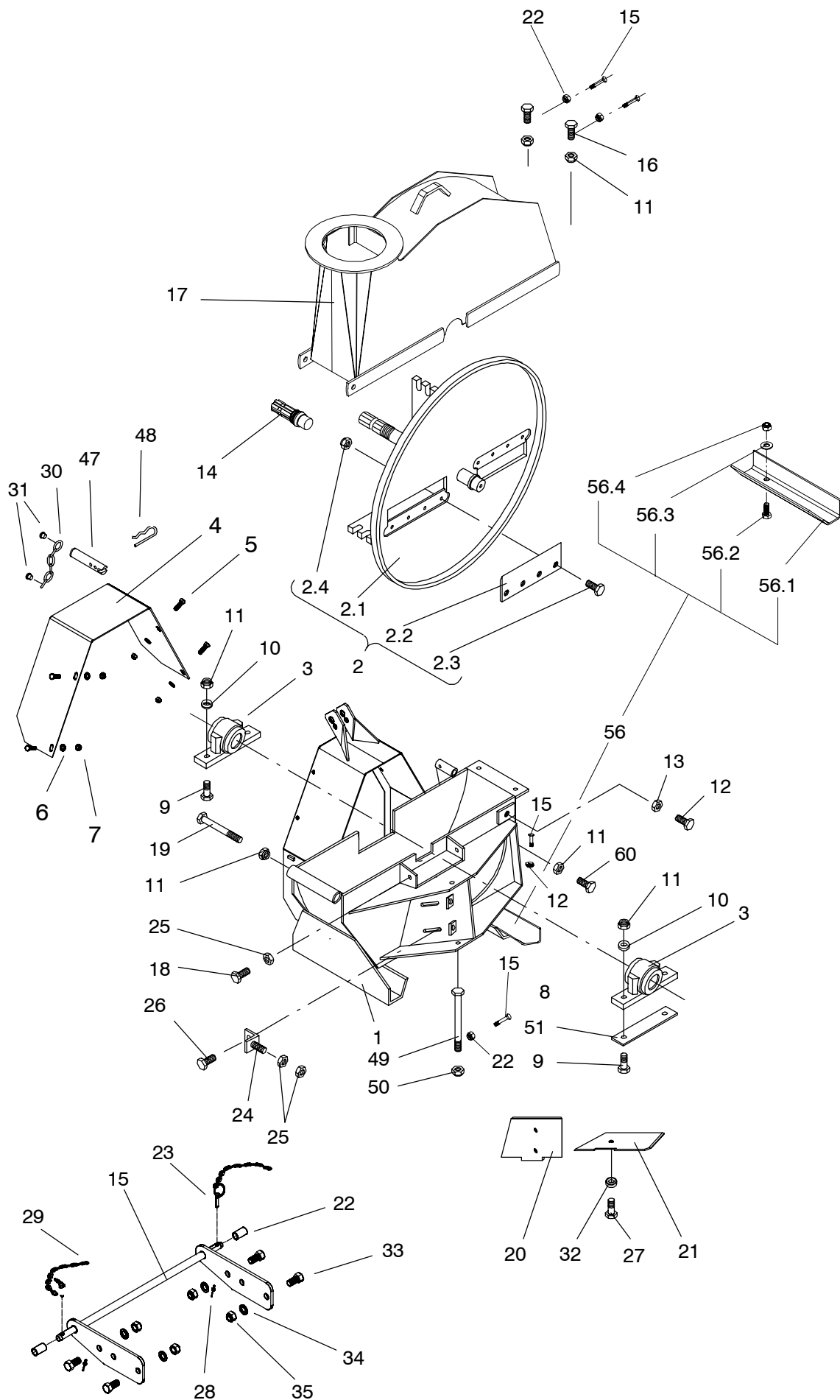


03514630

Part	Order no.	Description	Remarks	Qty.
.....	03514630	Discharge pipe, complete		
1.	43514600	Discharge pipe		1
2.	43510240	Lid		1
3.	52117108	Lock nut	M10 DIN985	4
4.	03510300	Chain	600	1
5.	94612082	Draw spring	26x2,6–106	1
6.	43340769	Fingernut		1
7.	43511670	Safety bolt	M16x35	1
8.	52840295	Cotter pin	5x24 DIN1481	3
9.	43340751	Reel		2
10. ...	52117082	Lock nut	M8 DIN985	1
11. ...	43511650	Safety screw	M10x50	2
12. ...	52200334	Washer	M8 DIN440	1

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CH140 L Chipper



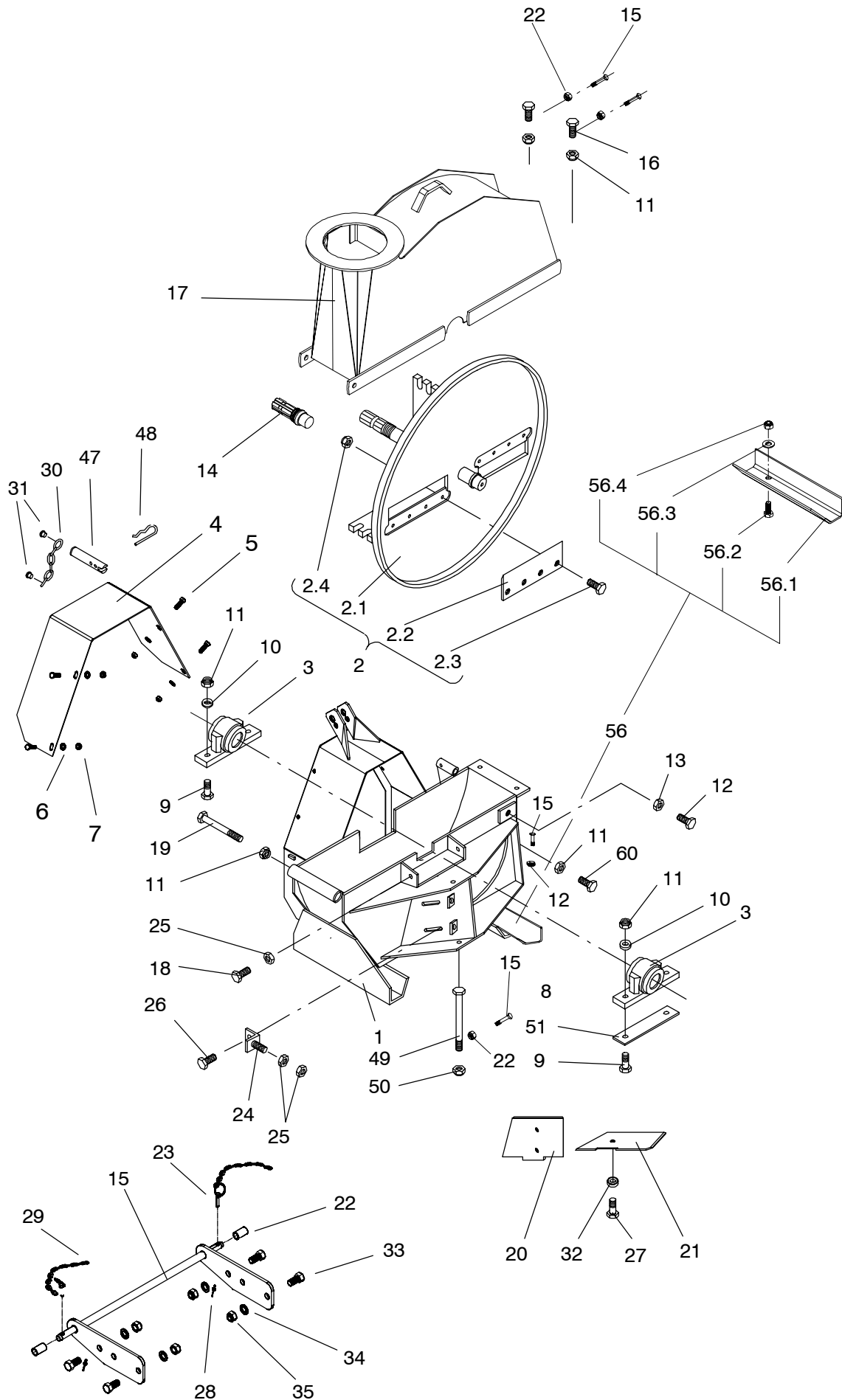
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Part	Order no.	Description	Remarks	Qty.
1. . . .	43345020	Lower housing		1
2. . . .	03510200	Disc, complete		1
2.1. . .	33510190	Disc		1
2.2. . .	43510220	Knife		2
2.3. . .	52060258	Screw	M10x40 DIN933	8
2.4. . .	52117108	Lock nut	M10 DIN985	8
3. . . .		Bearing	See separate spare parts list	
4. . . .	43345330	Plate		1
5. . . .	52060175	Screw	M8x25 DIN933	4
6. . . .	52200037	Washer	M8 DIN126	4
7. . . .	52117082	Lock nut	M8 DIN985	4
8. . . .	—			
9. . . .	52062056	Screw	M12x60 DIN931	4
10. . .	41215682	Washer		4
11. . .	52117124	Lock nut	M12 DIN985	7
12. . .	52062916	Screw	M16x60 DIN933	1
13. . .	52110079	Nut	M16 DIN934	1
14. . .	43510420	Shaft		1
15. . .	43341254	Drawbar		1
16. . .	43511630	Screw	M12x50	2
17. . .	33345150	Upper housing		1
18. . .	52062031	Screw	M12x40 DIN933	2
19. . .	43340546	Screw		1
20. . .	33341009	Vertical anvil		1
21. . .	33340993	Horizontal anvil		1
22. . .	40293797	Sleeve		2
23. . .	52842150	Ring pin	10x45	2
24. . .	43291103	Anvil adjuster		2
25. . .	52110053	Nut	M12 DIN934	8
26. . .	52062106	Screw	M16x30 DIN933	2
27. . .	52062098	Screw	M16x25 DIN933	1
28. . .	52832094	Rivet	4,8x15 Fe	2
29. . .	54826110	Chain	L=11 ¹³ / ₁₆ "	2
30. . .	03291143	Chain	250	1
31. . .	52832045	Rivet	4,8x19 Fe	2
32. . .	43310499	Washer		1
33. . .	52062213	Screw	M20x40 DIN933	4
34. . .	5211083	Spring washer	M20 DIN127	4
35. . .	52117207	Lock nut	DIN985	4

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CH140 L Chipper



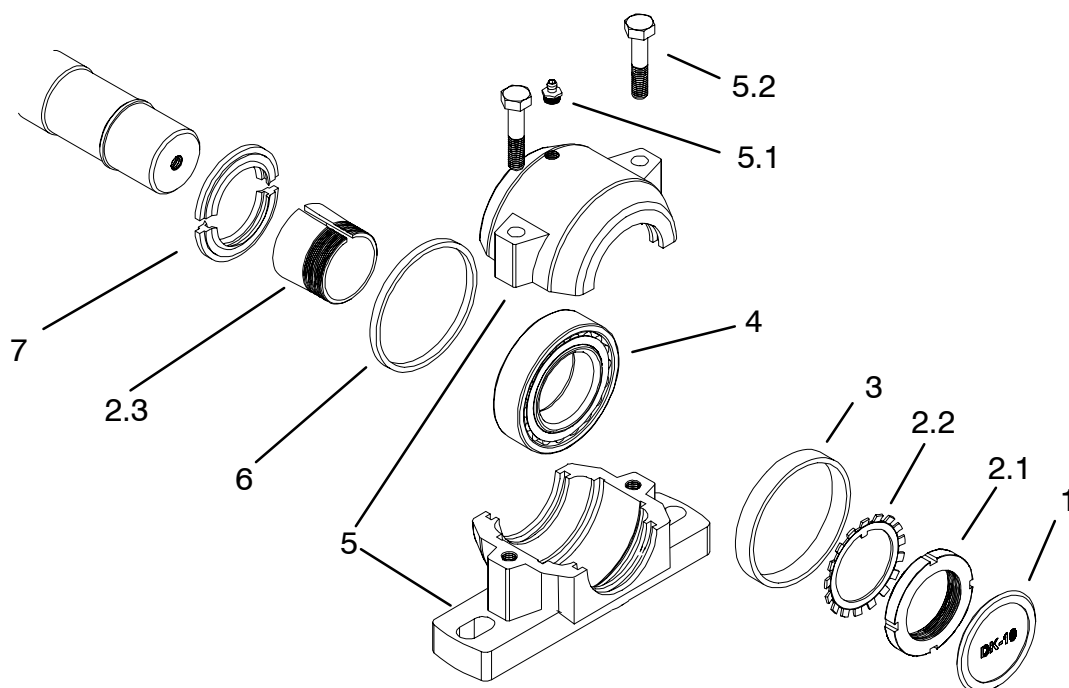
CH140 L

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Part	Order no.	Description	Remarks	Qty.
36.	..	—		
37.	..	—		
38.	..	—		
39.	..	—		
40.	..	—		
41.	.. 52060241	 Screw	M10x35 DIN933	1
42.	..	—		
43.	..	—		
44.	..	—		
45.	..	—		
46.	.. 52211042	 Spring washer	M10 DIN127	1
47.	.. 43341429	 Locking pin		1
48.	.. 52842143	 Cotter pin	5x105	1
49.	.. 43511660	 Safety screw	M16x240	1
50.	.. 52117165	 Lock nut	M16 DIN985	1
51.	.. 43341270	 Mounting plate		2
52.	..	—		
53.	..	—		
54.	..	—		
55.	..	—		
56.	.. 43513140	 Extension plate mounting		1
56.1	43512490	 Extension plate		1
56.2	52062023	 Screw	M12x30 DIN933	1
56.3	52200508	 washer	M12 DIN440	1
56.4	52117124	 Lock nut	M12 DIN985	1
57.	..	—		
58.	..	—		
59.	..	—		
60.	.. 43511630	 Safety screw	M12x50	2
61.	..			

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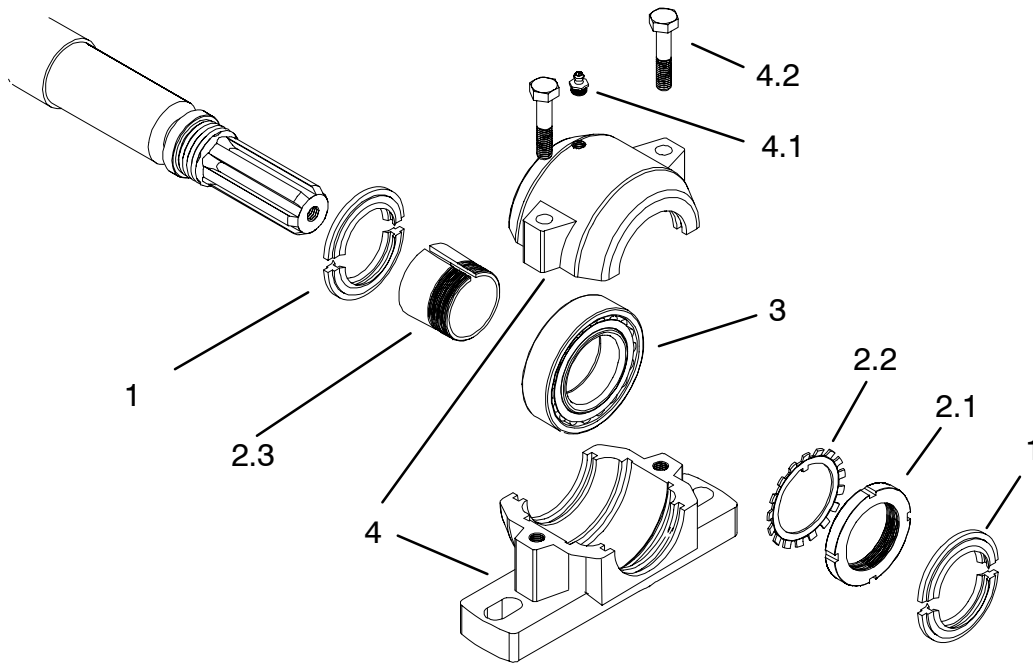
Bearing, feeder side



Part	Order no.	Description	Remarks	Qty.
1.	... 54513569	End plate		1
2.	... 54512371	Tightener sleeve, complete		1
2.1.		Shaft nut		1
2.2.		Lock plate		1
2.3.		Tightener sleeve		1
3.	... 43513360	Spacer ring D90x12,7		1
4.	... 54512363	Bearing		1
5.	... 54513590	Bearing housing		1
5.1.	... 52401015	Grease nipple	1/8"	1
5.2.		Screw	M10x50 DIN931 10.9	2
6.	... 43513350	Spacer ring D90x5,4		1
7.	... 52334232	Seal	Order no. including four pieces	1

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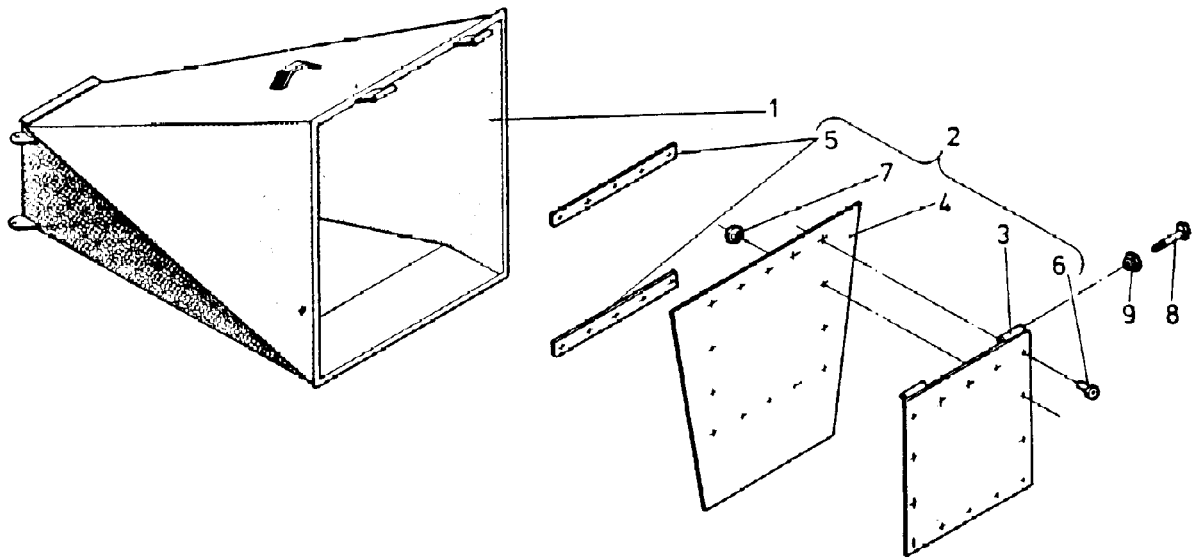
Bearing, splined axle side



Part	Order no.	Description	Remarks	Qty.
1. . . .	52334232	Seal	Order no. including four pieces	1
2. . . .	54512371	Tightener sleeve, complete		1
2.1.		Shaft nut		1
2.2.		Lock plate		1
2.3.		Tightener sleeve		1
3. . . .	54512363	Bearing		1
4. . . .	54513590	Bearing housing		1
4.1. . . .	52401015	Grease nipple	1/8"	1
4.2.		Screw	M10x50 DIN931 10.9	2

CH140 L

F150 Mechanical feed chute

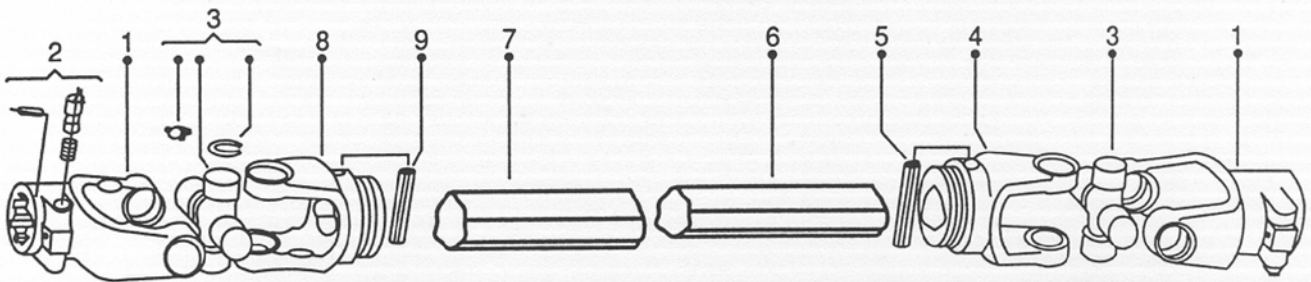


Part	Order no.	Description	Remarks	Qty.
1.	03342151	Feed chute		1
2.	43340876	Flap, complete	Including parts 3,4,5,6,7	1
3.	43340884	Flap		1
4.	43340900	Flap rubber		1
5.	43340918	Support		2
6.	52832094	Pop rivet	4,8x15	14
7.	52200011	Washer	M5 DIN126	4
8.	52060399	Bolt	M8x80 DIN931	2
9.	52117082	Nut	M8 DIN985	2

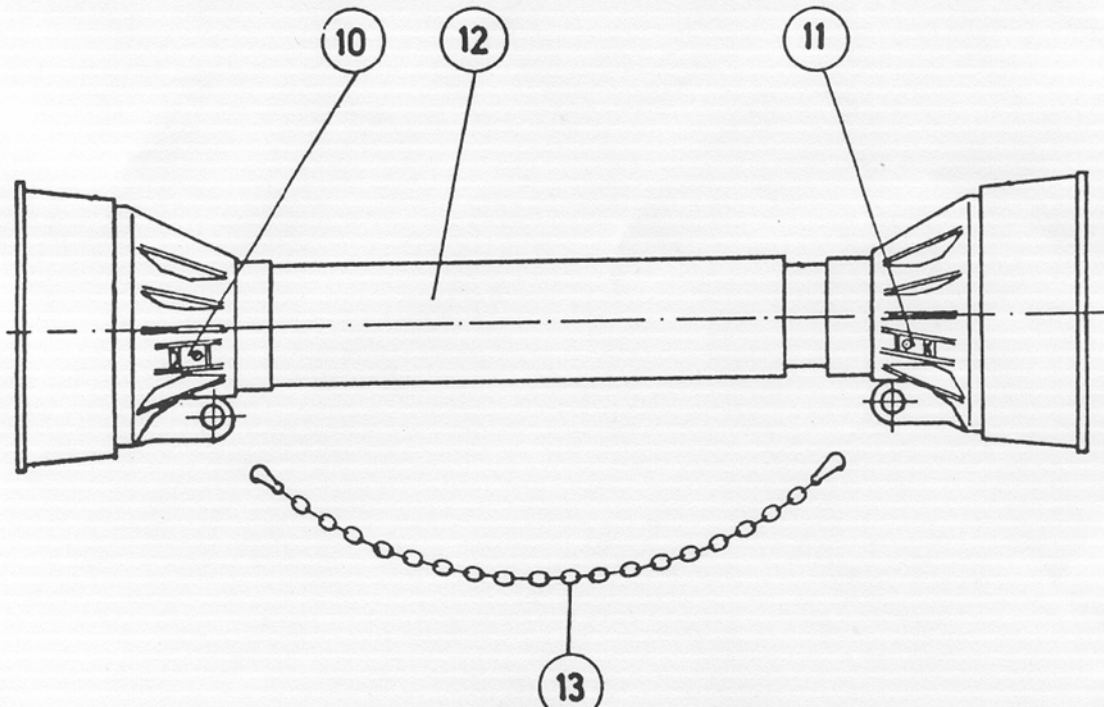
CH140 L

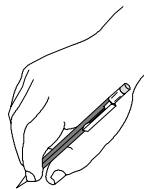
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M4



Pic	Part #	DESCRIPTION	Quan
1	8041	OUTER YOKE WITH QUICK COUPLER	2
2	1073	PUSH PIN SET	2
3	8081	CROSS	2
4	8115	INNER YOKE FOR INNER TUBE	1
5	4204	SPRING TENSION PIN FOR IT.	1
6	8106	INNER TUBE	1
7	8205	OUTER TUBE	1
8	8225	INNER YOKE FOR OUTER TUBE	1
9	8203	SPRING TENSION PIN FOR OT.	1
10	8263	PLASTIC RETAINING RING OT	1
11	8163	PLASTIC RETAINING RING IT	1
12	8091	SHIELD, COMPLETE	1
13	1000	CHAIN	1





A series of horizontal dashed lines spanning the width of the page, intended for handwriting practice. There are 20 lines in total, evenly spaced from the top of the writing area to the bottom.

LIMITED WARRANTY

FARMI Forestproducts are warranted to be free from defects in materials and workmanship for 12 months from the date of purchase. The warranty covers parts replacement only. The free replacement of parts is exclusive. The manufacturer is not liable for any incidental or consequential damages. No labor costs are included. This warranty does not cover defects resulting from:

- misuse of the product
- inadequate maintenance or
- modifications of the product

The cutting knives are not covered by the warranty.

The warranty is effective only, if the lower portion of this page is completed and returned to the manufacturer within 14 days of receipt of the product.

IMPORTANT:

Do not operate your machine without reading the operators manual! If the operators manual was lost, a new manual can be obtained from importer (write address below or call 607–589–6160).

NORTHEAST IMPLEMENT CORP.

**476 Halsey valley Road
SPENCER, NEW YORK 14883**



FARMI Forest
Ahmolantie 6
FIN–74 510 PELTOSALMI
FINLAND



Date of delivery: ____ / ____ 20 ____

Dealer: _____

Dealer's address: _____

Dealer's tel: _____

Product: _____

Serial number: _____



Return to the manufacturer

Date of delivery: ____ / ____ 20 ____

Dealer: _____

Dealer's address: _____

Dealer's tel: _____

Customer: _____

Customer's address: _____

Customer's tel: _____

Product: _____

Serial number: _____

Notes: _____



